

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for the main idea, the purpose, or the focus of the text, speech, or conversation.
- The topic can be used to organize, summarize, or evaluate the text, speech, or conversation.
- The topic can be related to other topics by finding similarities, differences, causes, effects, or implications.

Engineering Mathematics-I

Engineering Mathematics-I is a course that covers the basic concepts and techniques of calculus and its applications in engineering problems. The course aims to develop the students' ability to model, analyze, and solve engineering problems using mathematical tools. The course also introduces the students to the use of computer algebra systems for performing calculations and visualizing results.

The syllabus of Engineering Mathematics-I may vary depending on the university and the branch of engineering, but some of the common topics are:

- **Functions and Limits:** Definition and properties of functions, types of functions, limit and continuity of functions, L'Hospital's rule, indeterminate forms, asymptotes, and graphical representation of functions.
- **Differentiation and Applications:** Derivative of a function, rules of differentiation, chain rule, implicit differentiation, higher order derivatives, Leibniz's theorem, Rolle's theorem, mean value theorem, Taylor's theorem, maxima and minima of functions, curve sketching, and optimization problems.
- **Integration and Applications:** Indefinite and definite integrals, fundamental theorem of calculus, methods of integration, integration by parts, integration by substitution, integration by partial fractions, improper integrals, area under a curve, arc length, surface area, volume, and work.
- **Differential Equations and Applications:** Definition and classification of differential equations, order and degree of differential equations, solution of first order differential equations, separable equations, homogeneous equations, exact equations, linear equations, Bernoulli's equations, applications of differential equations, modelling of physical phenomena, population dynamics, radioactive decay, Newton's law of cooling, and orthogonal trajectories.
- **Vector Algebra and Applications:** Vectors and scalars, addition and subtraction of vectors, dot product and cross product of vectors, scalar

and vector triple products, applications of vector algebra, forces, moments, work, and torque.

Some of the learning outcomes of Engineering Mathematics-I are:

- To understand the concepts and principles of calculus and its relevance in engineering.
- To apply the techniques of differentiation and integration to solve engineering problems.
- To use differential equations to model and analyze engineering systems and phenomena.
- To perform vector operations and use them to describe physical quantities and situations.

Some of the references for Engineering Mathematics-I are:

- Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley, 2011.
- Grewal B.S., Higher Engineering Mathematics, 43rd Edition, Khanna Publishers, 2014.
- James Stewart, Calculus: Early Transcendentals, 8th Edition, Cengage Learning, 2016.
- R.K. Jain and S.R.K. Iyengar, Advanced Engineering Mathematics, 4th Edition, Narosa Publishing House, 2014.

Unit 1 - Matrices

A matrix is a rectangular array of numbers, symbols, or expressions, arranged in rows and columns. For example, the following is a matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

The dimensions of a matrix are the number of rows and columns it has. For example, the matrix above has 3 rows and 3 columns, so its dimensions are 3 x 3.

Some important types of matrices are:

- A square matrix is a matrix that has the same number of rows and columns. For example, the matrix above is a square matrix.
- A diagonal matrix is a square matrix that has nonzero entries only on the main diagonal (from the top left to the bottom right). For example, the following is a diagonal matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

- An identity matrix is a diagonal matrix that has 1s on the main diagonal and 0s elsewhere. For example, the following is an identity matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- A zero matrix is a matrix that has all entries equal to zero. For example, the following is a zero matrix:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- A row matrix is a matrix that has only one row. For example, the following is a row matrix:

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

- A column matrix is a matrix that has only one column. For example, the following is a column matrix:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

- A scalar matrix is a diagonal matrix that has the same nonzero entry on the main diagonal. For example, the following is a scalar matrix:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- A symmetric matrix is a square matrix that is equal to its transpose. The transpose of a matrix is obtained by swapping the rows and columns. For example, the following is a symmetric matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

- A skew-symmetric matrix is a square matrix that is equal to the negative of its transpose. For example, the following is a skew-symmetric matrix:

$$\begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}$$

Some basic operations on matrices are:

- Addition: Two matrices can be added if they have the same dimensions. The sum of two matrices is obtained by adding the corresponding entries. For example:

$$\begin{array}{|ccc|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ \hline 7 & 8 & 9 \\ \hline \end{array} + \begin{array}{|ccc|} \hline 4 & 5 & 6 \\ \hline 7 & 8 & 9 \\ \hline 1 & 2 & 3 \\ \hline \end{array} = \begin{array}{|ccc|} \hline 1+4 & 2+5 & 3+6 \\ \hline 4+7 & 5+8 & 6+9 \\ \hline 7+1 & 8+2 & 9+3 \\ \hline \end{array}$$

- Subtraction: Two matrices can be subtracted if they have the same dimensions. The difference of two matrices is obtained by subtracting the corresponding entries. For example:

$$\begin{array}{|ccc|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ \hline 7 & 8 & 9 \\ \hline \end{array} - \begin{array}{|ccc|} \hline 4 & 5 & 6 \\ \hline 7 & 8 & 9 \\ \hline 1 & 2 & 3 \\ \hline \end{array} = \begin{array}{|ccc|} \hline 1-4 & 2-5 & 3-6 \\ \hline 4-7 & 5-8 & 6-9 \\ \hline 7-1 & 8-2 & 9-3 \\ \hline \end{array}$$

- Scalar multiplication: A matrix can be multiplied by a scalar (a number). The product of a scalar and a matrix is obtained by multiplying each entry of the matrix by the scalar. For example:

$$2 \times \begin{array}{|cc|} \hline 1 & 2 \\ \hline \end{array}$$

Elementary transformations for the notes of the Unit 1 - Matrices in the subject of ENGINEERING MATHEMATICS-I

- Elementary transformations are operations done on the rows and columns of matrices to change their shape so that the computations become easier.
- Elementary transformations are also used to discover the inverse of a matrix, the determinants of a matrix, and to solve a system of linear equations.
- There are three types of elementary transformations for matrices :
 - Interchanging two rows or two columns of a matrix. For example, if $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then interchanging the first and second rows gives $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$.
 - Multiplying a row or a column of a matrix by a non-zero scalar. For example, if $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then multiplying the first row by 2 gives $A = \begin{bmatrix} 2 & 4 \\ 3 & 4 \end{bmatrix}$.
 - Adding a multiple of one row or one column to another row or another column of a matrix. For example, if $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then adding the first row to the second row gives $A = \begin{bmatrix} 1 & 2 \\ 4 & 6 \end{bmatrix}$.
- An elementary matrix is a square matrix that is obtained by applying one elementary transformation to the identity matrix. For example, if $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then E is an elementary matrix that is obtained by interchanging the first and second rows of the 2×2 identity matrix $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- The elementary matrices generate the general linear group $GL_n(F)$ when F is a field. Left multiplication (pre-multiplication) by an elementary matrix represents elementary row operations, while right multiplication (post-multiplication) represents elementary column operations.

- The inverse of an elementary matrix is also an elementary matrix of the same type. For example, if $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then $E^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ as well.

Inverse of a matrix

- The inverse of a square matrix A is another matrix A^{-1} such that $A * A^{-1} = A^{-1} * A = I$, where I is the identity matrix .
- A square matrix has an inverse if and only if its determinant is non-zero . A square matrix that is not invertible is called singular or degenerate.
- The inverse of a matrix is used to find the solution of linear equations through the matrix inversion method.
- There are different methods to find the inverse of a matrix, such as the adjoint method, the Gauss-Jordan method, and the row reduction method.
- Some properties of inverse matrices are:
 - The inverse of inverse matrix is equal to the original matrix, i.e., $(A^{-1})^{-1} = A$.
 - If A and B are invertible matrices, then AB is also invertible, and $(AB)^{-1} = B^{-1}A^{-1}$.
 - If A is nonsingular, then $(A^T)^{-1} = (A^{-1})^T$, where A^T is the transpose of A .

Rank of Matrix

- The rank of a matrix is a measure of the linear independence of its rows or columns.
- The rank of a matrix is equal to the number of linearly independent rows (or columns) in it .
- The rank of a matrix cannot be more than its number of rows and columns .
- The rank of a matrix is also equal to the order of the largest non-zero minor in it . A minor is a determinant of a square submatrix of the original matrix.
- A matrix is said to have full rank if its rank equals the largest possible for a matrix of the same dimensions, which is the lesser of the number of rows and columns.
- A matrix is said to be rank-deficient if it does not have full rank. The rank deficiency of a matrix is the difference between the lesser of the number of rows and columns, and the rank.
- The rank of a matrix can be found by using various methods, such as row reduction, Gaussian elimination, or finding the non-zero eigenvalues of the matrix.
- The rank of a matrix has many applications in linear algebra, such as solving systems of linear equations, finding the inverse of a matrix, determining the linear dependence or independence of vectors, and computing

the determinant of a matrix.

Solution of system of linear equations

- A system of linear equations is a set of equations that involve the same variables and can be written in the form of $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$, where $a_1, a_2, \dots, a_n$ and b are constants and $x_1, x_2, \dots, x_n$ are variables.
- A solution to a system of linear equations is an assignment of values to the variables such that all the equations are simultaneously satisfied. For example, the ordered pair $(4, 7)$ is a solution to the system of linear equations $x + y = 11$ and $2x - y = 1$.
- A system of linear equations can have one, infinitely many, or no solutions, depending on the relationship between the equations. For example, the system of linear equations $x + y = 5$ and $x - y = 3$ has one solution $(4, 1)$, the system of linear equations $x + y = 5$ and $2x + 2y = 10$ has infinitely many solutions $(x, 5 - x)$, and the system of linear equations $x + y = 5$ and $x + y = 7$ has no solution.
- There are multiple methods of solving systems of linear equations, such as graphing, substitution, elimination, matrix methods, and Cramer's rule. Each method has its own advantages and disadvantages, depending on the type and number of equations, the number of variables, and the coefficients of the equations.
- Graphing method: This method involves plotting the graphs of the equations on the same coordinate plane and finding the point of intersection, if any, which is the solution to the system. This method is useful for visualizing the relationship between the equations, but it may not be very accurate or efficient for large or complicated systems.
- Substitution method: This method involves solving one equation for one variable in terms of the others, and then substituting the expression into the other equations to eliminate that variable. This process is repeated until only one variable remains, which can then be solved. This method is useful for systems with two or three equations and variables, but it may become tedious or difficult for larger or more complex systems.
- Elimination method: This method involves adding or subtracting multiples of the equations to eliminate one variable at a time, until only one variable remains, which can then be solved. This method is useful for systems with two or three equations and variables, but it may require careful manipulation of the coefficients to avoid fractions or large numbers.
- Matrix methods: This method involves representing the system of linear equations as a matrix equation of the form $AX = B$, where A is the coefficient matrix, X is the variable matrix, and B is the constant matrix. This method is useful for systems with any number of equations and variables, but it requires knowledge of matrix operations and properties, such as inverse, determinant, and rank.

- Cramer's rule: This method involves finding the solution to the system of linear equations by using the determinants of the coefficient matrix and the matrices obtained by replacing one column of the coefficient matrix with the constant matrix. This method is useful for systems with any number of equations and variables, but it requires computing the determinants of large matrices, which can be time-consuming or error-prone.

Characteristic equation

- The characteristic equation is the equation that is used to find the eigenvalues of a square matrix.
- The eigenvalues are the scalars that satisfy the equation $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, where $\mathbf{A}$ is the matrix and $\mathbf{v}$ is a nonzero vector called an eigenvector.
- The characteristic equation is obtained by subtracting λ from the diagonal entries of $\mathbf{A}$ and then finding the determinant of the resulting matrix. This determinant is called the characteristic polynomial of $\mathbf{A}$.
- The characteristic equation can be written as $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$, where $\mathbf{I}$ is the identity matrix of the same size as $\mathbf{A}$.
- The characteristic equation is a polynomial equation in λ of degree n , where n is the order of $\mathbf{A}$. The roots of this equation are the eigenvalues of $\mathbf{A}$.
- The characteristic equation is invariant under matrix similarity, meaning that if $\mathbf{A}$ and $\mathbf{B}$ are similar matrices, then they have the same characteristic equation and the same eigenvalues.
- The characteristic equation can be used to find the properties of a matrix, such as its rank, trace, determinant, and inverse. It can also be used to solve systems of linear differential equations, linear recurrence relations, and matrix differential equations.

Cayley-Hamilton Theorem and its application

- The Cayley-Hamilton theorem is a result in linear algebra that states that every square matrix satisfies its own characteristic equation.
- The characteristic equation of a square matrix A is obtained by equating the determinant of $A - xI$ to zero, where x is a scalar variable and I is the identity matrix of the same size as A .
- The characteristic equation is a polynomial in x , called the characteristic polynomial of A , and has the same degree as the size of A .
- The Cayley-Hamilton theorem says that if we substitute A for x in the characteristic polynomial, we get the zero matrix.
- For example, if A is a 2×2 matrix with entries a , b , c , and d , then its characteristic polynomial is $p(x) = x^2 - (a + d)x + (ad - bc)$, and the Cayley-Hamilton theorem says that $p(A) = A^2 - (a + d)A + (ad - bc)I = 0$.
- The Cayley-Hamilton theorem can be used to find the inverse of a matrix,

if it exists, by expressing A^{-1} as a linear combination of powers of A using the characteristic polynomial.

- For example, if A is a 2×2 matrix with entries a, b, c , and d , and $ad - bc$ is not zero, then its inverse is $A^{-1} = (1/(ad - bc))(dI - bA + cA^2)$.
- The Cayley-Hamilton theorem can also be used to find the minimal polynomial of a matrix, which is the monic polynomial of the smallest degree that annihilates the matrix.
- The minimal polynomial of a matrix always divides its characteristic polynomial, and they are equal if and only if the matrix has distinct eigenvalues.
- The Cayley-Hamilton theorem has applications in various fields of mathematics, such as control theory, commutative algebra, and algebraic number theory.

Linear Dependence and Independence of Vectors

- A vector is a quantity that has both magnitude and direction. Examples of vectors are displacement, velocity, force, etc.
- A vector can be represented by an arrow or by a column matrix of its components.
- A linear combination of vectors is an expression of the form $a_1v_1 + a_2v_2 + \dots + a_nv_n$, where $a_1, a_2, \dots, a_n$ are scalars and $v_1, v_2, \dots, v_n$ are vectors.
- A set of vectors is said to be linearly dependent if there exists a nontrivial linear combination of them that equals the zero vector. That is, if there exist scalars $a_1, a_2, \dots, a_n$, not all zero, such that $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$.
- A set of vectors is said to be linearly independent if the only linear combination of them that equals the zero vector is the trivial one, where all the scalars are zero. That is, if $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$ implies that $a_1 = a_2 = \dots = a_n = 0$.
- Linear dependence and independence are properties of a set of vectors, not of individual vectors. A single vector is always linearly dependent on itself, but it is linearly independent from any other vector.
- A set of vectors that contains the zero vector is always linearly dependent, since any nonzero scalar times the zero vector gives the zero vector.
- To check if a set of vectors is linearly dependent or independent, one can use the following methods:
 - Write the vectors as column matrices and form a matrix by placing them side by side. Then perform row operations to reduce the matrix to its row echelon form. If the row echelon form has a row of zeros, then the set of vectors is linearly dependent. Otherwise, it is linearly independent.
 - Write the vectors as column matrices and form a matrix by placing them side by side. Then find the determinant of the matrix. If the determinant is zero, then the set of vectors is linearly dependent.

- Otherwise, it is linearly independent.
- Write the linear combination $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$ as a system of linear equations in the scalars $a_1, a_2, \dots, a_n$. Then solve the system using any method. If the system has a nontrivial solution, then the set of vectors is linearly dependent. Otherwise, it is linearly independent.
 - Linear dependence and independence are important concepts in linear algebra, as they are related to the notions of span, basis, dimension, rank, nullity, etc.

Eigenvalues and Eigenvectors

- Eigenvalues and eigenvectors are concepts related to linear transformations of vector spaces.
- A linear transformation is a function that maps vectors to vectors, such that the sum and scalar multiplication of vectors are preserved.
- A matrix is a rectangular array of numbers that can represent a linear transformation by multiplying it with a vector.
- An eigenvector of a matrix is a nonzero vector that does not change its direction when multiplied by the matrix. It may only change its length or sign.
- An eigenvalue of a matrix is a scalar that corresponds to an eigenvector. It is the factor by which the eigenvector is scaled when multiplied by the matrix.
- Geometrically, an eigenvector points in a direction that is invariant under the linear transformation, and the eigenvalue is the amount of stretching or shrinking in that direction.
- Mathematically, an eigenvector $\mathbf{x}$ and an eigenvalue λ of a matrix $\mathbf{A}$ satisfy the equation $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$.
- To find the eigenvalues and eigenvectors of a matrix, one can solve the characteristic equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$, where $\mathbf{I}$ is the identity matrix and $\det$ is the determinant function.
- The characteristic equation is a polynomial of degree n , where n is the size of the matrix. Therefore, there are at most n distinct eigenvalues for a matrix.
- The eigenvectors corresponding to a given eigenvalue can be found by solving the system of linear equations $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$.
- Eigenvectors corresponding to different eigenvalues are linearly independent, meaning they cannot be expressed as a linear combination of each other.

- Some properties of eigenvalues and eigenvectors are:
 - If $\mathbf{A}$ is a triangular matrix, then the diagonal elements of $\mathbf{A}$ are the eigenvalues of $\mathbf{A}$.
 - If λ is an eigenvalue of $\mathbf{A}$ with eigenvector $\mathbf{x}$, then $1/\lambda$ is an eigenvalue of $\mathbf{A}^{-1}$ with eigenvector $\mathbf{x}$.
 - If λ is an eigenvalue of $\mathbf{A}$, then λ is an eigenvalue of $\mathbf{A}^T$, where T denotes the transpose operation.
 - The sum of the eigenvalues of $\mathbf{A}$ is equal to the trace of $\mathbf{A}$, which is the sum of the diagonal elements of $\mathbf{A}$.
 - The product of the eigenvalues of $\mathbf{A}$ is equal to the determinant of $\mathbf{A}$.
 - If $\mathbf{A}$ and $\mathbf{B}$ are similar matrices, meaning they are related by $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$ for some invertible matrix $\mathbf{P}$, then they have the same eigenvalues.
 - If $\mathbf{A}$ is symmetric, meaning $\mathbf{A} = \mathbf{A}^T$, then its eigenvalues are real and its eigenvectors are orthogonal, meaning they are perpendicular to each other.
 - If $\mathbf{A}$ is positive definite, meaning $\mathbf{x}^T\mathbf{A}\mathbf{x} > 0$ for any nonzero vector $\mathbf{x}$, then its eigenvalues are positive and its eigenvectors are linearly independent.
- Eigenvalues and eigenvectors have many applications in mathematics, physics, engineering, and other fields. They can be used to:
 - Diagonalize a matrix, meaning to find a matrix $\mathbf{D}$ that is diagonal and similar to $\mathbf{A}$. This simplifies the computation of matrix powers and exponentials.
 - Decompose a matrix, meaning to find a matrix $\mathbf{P}$ that is orthogonal and a matrix $\mathbf{D}$ that is diagonal, such that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T$. This reveals the geometric structure of the linear transformation.
 - Find the principal components of a data set, meaning to find the directions that capture the most variance in the data. This reduces the dimensionality and noise of the data.
 - Solve differential equations, meaning to find the solutions of equations that involve derivatives of functions. This can model the behavior of physical systems over time.
 - Perform spectral analysis, meaning to find the frequencies and amplitudes of periodic signals. This can identify the components and characteristics of the signals.

Complex Matrices

- A complex matrix is a matrix that has some complex number among its elements.
- A complex number is a number made up of a real part and an imaginary

part, which is indicated by the letter i .

- For example, the matrix

$$A = \begin{bmatrix} 1 + 2i & -3i \\ 4 - i & 2 + 5i \end{bmatrix}$$

is a complex matrix.

- The set of all $m \times n$ complex matrices is denoted as $\mathbb{C}^{m \times n}$, or complex $m \times n$ matrices.
- Complex matrices have similar properties and operations as real matrices, such as addition, subtraction, multiplication, transpose, inverse, determinant, rank, etc.
- However, some operations require a modification when dealing with complex matrices, such as the dot product, the conjugate, the norm, the eigenvalues, and the eigenvectors.
- The dot product of two complex vectors is defined as

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i \overline{v_i}$$

where $\overline{v_i}$ is the complex conjugate of v_i . - The complex conjugate of a complex number is obtained by changing the sign of the imaginary part, such as $\overline{1 + 2i} = 1 - 2i$. - The complex conjugate of a complex matrix is a matrix whose entries are the complex conjugate of the entries of the original matrix, such as

$$\overline{A} = \begin{bmatrix} 1 - 2i & 3i \\ 4 + i & 2 - 5i \end{bmatrix}$$

- The norm of a complex vector is defined as

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{\sum_{i=1}^n |u_i|^2}$$

where $|u_i|$ is the modulus of u_i , which is the distance from the origin to the point representing u_i in the complex plane. - The eigenvalues and eigenvectors of a complex matrix are complex numbers and vectors that satisfy the equation

$$A\mathbf{x} = \lambda\mathbf{x}$$

where A is a complex matrix, $\mathbf{x}$ is a nonzero complex vector, and λ is a complex number. - The eigenvalues of a complex matrix can be found by solving the characteristic equation

$$\det(A - \lambda I) = 0$$

where I is the identity matrix of the same size as A . - The eigenvectors of a complex matrix can be found by plugging in the eigenvalues into the equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

and solving for $\mathbf{x}$.

Hermitian Matrix

- A **Hermitian matrix** is a complex square matrix that is equal to its own conjugate transpose .
- The **conjugate transpose** of a matrix is obtained by taking the complex conjugate of each element and then transposing the matrix.
- The **complex conjugate** of a complex number is obtained by changing the sign of the imaginary part.
- For example, if A is a complex matrix, then its conjugate transpose is denoted by A^H and defined as $A^H = (A^*)^T$, where A^* is the complex conjugate of A and A^T is the transpose of A .
- A Hermitian matrix satisfies the following properties :
 - It is **square**, meaning it has the same number of rows and columns.
 - It has **real** diagonal elements, meaning the elements on the main diagonal have zero imaginary parts.
 - It is **symmetric**, meaning the elements below and above the main diagonal are mirror images of each other.
 - It is **conjugate symmetric**, meaning the element in the i -th row and j -th column is the complex conjugate of the element in the j -th row and i -th column, for all indices i and j .
 - It has **real eigenvalues**, meaning the solutions of the characteristic equation are all real numbers.
 - It has **orthogonal eigenvectors**, meaning the eigenvectors corresponding to distinct eigenvalues are perpendicular to each other.
- A Hermitian matrix can be written in the form $A = B + iC$, where B and C are real symmetric matrices.
- A Hermitian matrix can also be written in the form $A = UDU^H$, where U is a unitary matrix and D is a diagonal matrix with real entries.
- A Hermitian matrix is a special case of a **normal matrix**, which is a matrix that commutes with its conjugate transpose, i.e., $A^H A = A A^H$.
- A Hermitian matrix is also a special case of a **self-adjoint operator**, which is a linear transformation that is equal to its adjoint, i.e., $A = A^*$, where A^* is the adjoint of A .
- A Hermitian matrix is also a special case of a **positive-definite matrix**, which is a matrix that satisfies $\mathbf{x}^H A \mathbf{x} > 0$ for all nonzero vectors $\mathbf{x}$.

- A Hermitian matrix is also a special case of a **Hermitian form**, which is a bilinear form that is conjugate symmetric, i.e., $f(x, y) = \overline{f(y, x)}$ for all vectors x and y .
- A Hermitian matrix is also a special case of a **Hermitian inner product**, which is a function that assigns a complex number to every pair of vectors, satisfying the following properties:
 - Conjugate symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$ for all vectors x and y .
 - Linearity: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$ for all scalars a and b and vectors x, y and z .
 - Positive-definiteness: $\langle x, x \rangle \geq 0$ for all vectors x and $\langle x, x \rangle = 0$ if and only if $x = 0$.
- A Hermitian matrix is also a special case of a **Hermitian manifold**, which is a smooth manifold with a Hermitian inner product on each tangent space.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is some information on skew-Hermitian matrices for the notes of the Unit 1 - Matrices in the subject of ENGINEERING MATHEMATICS-I.

Skew-Hermitian Matrices

Definition

A square matrix A is called **skew-Hermitian** (or **antihermitian**) if it satisfies the following property:

$$A^* = -A$$

where A^* is the **conjugate transpose** of A , which means the matrix obtained by taking the complex conjugate of each element of A and then transposing the matrix.

Properties

Some of the properties of skew-Hermitian matrices are:

- The diagonal elements of a skew-Hermitian matrix are either zeros or pure imaginary numbers (i.e., numbers of the form bi , where b is a real number and i is the imaginary unit).
- A skew-Hermitian matrix is an example of a **normal matrix**, which means that it commutes with its conjugate transpose, i.e., $AA^* = A^*A$.
- A skew-Hermitian matrix is **diagonalizable**, which means that it can be written as $A = PDP^{-1}$, where P is a unitary matrix (i.e., $P^* = P^{-1}$) and D is a diagonal matrix.
- The eigenvalues of a skew-Hermitian matrix are either purely imaginary or zeros. Furthermore, the eigenvectors of a skew-Hermitian matrix corresponding to distinct eigenvalues are orthogonal.

Examples

Here are some examples of skew-Hermitian matrices:

- $A = \begin{bmatrix} 0 & i & -i & 0 \end{bmatrix}$ is a 2×2 skew-Hermitian matrix, since $A^* = \begin{bmatrix} 0 & -i & i & 0 \end{bmatrix} = -A$. Its eigenvalues are $\pm i$ and its eigenvectors are $\begin{bmatrix} 1 & i \end{bmatrix}$ and $\begin{bmatrix} 1 & -i \end{bmatrix}$.
- $B = \begin{bmatrix} 0 & 1 + i & 2 - 3i & -1 - i & 0 & 4 + i & -2 + 3i & -4 - i & 0 \end{bmatrix}$ is a 3×3 skew-Hermitian matrix, since $B^* = \begin{bmatrix} 0 & -1 - i & -2 + 3i & 1 + i & 0 & -4 - i & 2 - 3i & 4 + i & 0 \end{bmatrix} = -B$. Its eigenvalues are $0, \pm 2i$ and its eigenvectors are $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & -i & 1 + i \end{bmatrix}$ and $\begin{bmatrix} 1 & i & 1 - i \end{bmatrix}$.

Unitary Matrices

A unitary matrix is a complex square matrix that satisfies the following equation:

$$U^H U = U U^H = I$$

where U^H is the conjugate transpose of U and I is the identity matrix. This means that the inverse of a unitary matrix is its conjugate transpose.

Some examples of unitary matrices are:

$$U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{6} & -1/\sqrt{6} & 2/\sqrt{6} \\ -1/\sqrt{2} & 0 & 0 \end{bmatrix}$$

$$W = \begin{bmatrix} 1 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{bmatrix}$$

Some properties of unitary matrices are:

- The unitary matrix is a non-singular matrix.
- The unitary matrix is an invertible matrix.
- The product of two unitary matrices is a unitary matrix.
- The inverse of a unitary matrix is another unitary matrix.
- A matrix is unitary, if and only if its transpose is unitary.
- A matrix is unitary if its rows are orthonormal, and the columns are orthonormal.
- The determinant of a unitary matrix has absolute value 1.
- The eigenvalues of a unitary matrix have absolute value 1.
- The trace of a unitary matrix is the sum of its eigenvalues.
- The unitary matrices form a group under matrix multiplication, called the unitary group.

Applications of Matrices in Engineering Problems

Matrices are rectangular arrays of numbers that can represent data, equations, transformations, and systems of linear equations. Matrices have many applica-

tions in engineering, such as:

- Solving electrical circuits using Kirchhoff's laws and matrix inversion .
- Encrypting and decrypting messages using matrix multiplication and inverse.
- Modeling vibrations, rotations, and oscillations using eigenvalues and eigenvectors .
- Analyzing graphs, networks, and algorithms using adjacency matrices and incidence matrices.
- Performing linear transformations, such as scaling, rotation, reflection, and projection using transformation matrices .
- Simulating quantum mechanics, optics, and relativity using matrices and tensors .
- Computing statistics and probability using matrices and determinants .
- Generating solar energy using matrices and vectors of sun .

Matrices are useful tools for engineering problems because they can simplify complex calculations, organize data, and manipulate variables. Matrices can also be used to approximate nonlinear functions and systems using linearization techniques.

Unit 2 - Differential Calculus- I

- Differential calculus is the branch of mathematics that studies the rates of change of functions and their properties.
- The main concept of differential calculus is the derivative, which measures the instantaneous rate of change of a function at a point.
- The derivative of a function $f(x)$ is denoted by $f'(x)$ or dy/dx , where $y = f(x)$.
- The derivative of a function $f(x)$ can be interpreted as the slope of the tangent line to the graph of $f(x)$ at a point x , or as the limit of the ratio of the change in $f(x)$ to the change in x as x approaches a point.
- The derivative of a function $f(x)$ can be calculated using various rules and formulas, such as the power rule, the product rule, the quotient rule, the chain rule, and the implicit differentiation.
- The derivative of a function $f(x)$ can be used to find the critical points, the extrema, the intervals of increase and decrease, the concavity, the inflection points, and the asymptotes of the graph of $f(x)$.
- The derivative of a function $f(x)$ can also be used to solve various problems involving optimization, related rates, linear approximation, differentials, and Newton's method.
- Some important applications of differential calculus are in physics, engineering, economics, biology, and chemistry, where the rates of change of various quantities are of interest.

Successive Differentiation (nth order derivatives)

- Successive differentiation is the process of differentiating a given function successively n times and the results of such differentiation are called successive derivatives.
- The n th derivative of a function $f(x)$ is denoted by $f^{(n)}(x)$ or $D^n f(x)$ or $y^{(n)}$ where $y = f(x)$.
- The n th derivative of a function $f(x)$ is the first-order derivative of the $(n-1)$ th derivative of $f(x)$, i.e., $f^{(n)}(x) = (f^{(n-1)}(x))'$ or $D^n f(x) = D(D^{(n-1)} f(x))$.
- The n th derivative of a function $f(x)$ can be obtained by applying the chain rule, the product rule, the quotient rule, or the Leibnitz theorem, depending on the form of the function .
- The Leibnitz theorem states that the n th derivative of the product of two functions $u(x)$ and $v(x)$ is given by:

$$f^{(n)}(x) = (u(x)v(x))^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(n-k)}(x) v^{(k)}(x)$$

where $\binom{n}{k}$ is the binomial coefficient .

- Some properties of the n th derivative are:
 - If $f(x)$ is a polynomial of degree n , then $f^{(n+1)}(x) = 0$ for all x .
 - If $f(x)$ is a constant function, then $f^{(n)}(x) = 0$ for all $n > 0$ and x .
 - If $f(x)$ is an exponential function of the form a^x , then $f^{(n)}(x) = (\ln a)^n a^x$ for all n and x .
 - If $f(x)$ is a trigonometric function of the form $\sin(ax)$ or $\cos(ax)$, then $f^{(n)}(x) = a^n \sin(ax + n\pi/2)$ or $a^n \cos(ax + n\pi/2)$ respectively, for all n and x .

Leibnitz Theorem

- Leibnitz theorem is a generalization of the product rule of differentiation .
- It states that if the functions $u(x)$ and $v(x)$ are differentiable n times, then their product $u(x).v(x)$ is also differentiable n times .
- The formula for the n th derivative of the product of two functions is given by :

$$(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(n-k)} v^{(k)}$$

- Where $\binom{n}{k}$ is the binomial coefficient, and $u^{(n-k)}$ and $v^{(k)}$ are the $(n-k)$ th and k th derivatives of u and v respectively .
- The proof of this theorem is based on induction and the product rule of differentiation .

- The theorem can be extended to the case where the limits of integration are functions of x , and the integrand is a function of both x and t . This is known as the Leibniz integral rule or the differentiation under the integral sign.
- The formula for the Leibniz integral rule is given by:

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, t) dt = f(x, b(x)) \frac{db}{dx} - f(x, a(x)) \frac{da}{dx} + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x, t) dt$$

- Where $a(x)$ and $b(x)$ are the lower and upper limits of integration, and $f(x, t)$ is the integrand.
- The proof of this rule is based on the fundamental theorem of calculus and the chain rule of differentiation.

Curve tracing for the notes of the Unit 2 - Differential Calculus- I in the subject of ENGINEERING MATHEMATICS-I

- Curve tracing is the method of studying the properties and shape of a curve whose equation is given in cartesian, polar or parametric form.
- Curve tracing helps to sketch the graph of a function using the information obtained from its derivatives, such as extrema, concavity, inflection points, asymptotes, etc .
- Curve tracing also helps to find the traces of a surface, which are the curves that represent the intersection of the surface and a plane.

Steps for curve tracing

- Identify the type of equation (cartesian, polar or parametric) and the degree of the curve (linear, quadratic, cubic, etc).
- Find the domain and range of the function, and the intervals where it is defined, continuous and differentiable.
- Find the x - and y -intercepts of the curve, if any, by setting $y = 0$ and $x = 0$ respectively.
- Find the symmetry of the curve, if any, by checking if $f(x) = f(-x)$, $f(-x) = -f(x)$, or $f(x) = -f(-x)$.
- Find the asymptotes of the curve, if any, by checking the limits of the function as x or y approaches infinity or a finite value.
- Find the first derivative of the function and use it to find the critical points, where $f'(x) = 0$ or undefined, and the intervals where the function is increasing or decreasing.
- Find the second derivative of the function and use it to find the points of inflection, where $f''(x) = 0$ or undefined, and the intervals where the function is concave up or concave down.

- Plot the points and intervals obtained from the previous steps on a number line and use them to sketch the curve.
- Label the important features of the curve, such as intercepts, extrema, inflection points, asymptotes, etc.

Example of curve tracing

- Consider the function $f(x) = x^3 - 3x^2 - 9x + 5$.
- The equation is cartesian and the degree is cubic.
- The domain of the function is all real numbers, and the range is also all real numbers. The function is defined, continuous and differentiable for all x .
- The x -intercepts are the solutions of $f(x) = 0$, which are $x = -1$, $x = 1$, and $x = 5$. The y -intercept is $f(0) = 5$.
- The function has no symmetry, as $f(x) \neq f(-x)$, $f(-x) \neq -f(x)$, and $f(x) \neq -f(-x)$.
- The function has no asymptotes, as the limits of the function as x or y approaches infinity or a finite value are either finite or do not exist.
- The first derivative of the function is $f'(x) = 3x^2 - 6x - 9$. The critical points are the solutions of $f'(x) = 0$, which are $x = -1$ and $x = 3$. The function is increasing on $(-\infty, -1)$ and $(3, \infty)$, and decreasing on $(-1, 3)$.
- The second derivative of the function is $f''(x) = 6x - 6$. The point of inflection is the solution of $f''(x) = 0$, which is $x = 1$. The function is concave up on $(1, \infty)$, and concave down on $(-\infty, 1)$.
- The sketch of the curve is shown below, with the important features labeled.

curve sketch

Partial derivatives

- A partial derivative is a derivative where we hold some variables constant and find the rate of change of a function with respect to one variable .
- For example, if $f(x,y)$ is a function of two variables, then the partial derivative of f with respect to x is denoted by f_x or f / x and is obtained by treating y as a constant and differentiating f with respect to x .
- Similarly, the partial derivative of f with respect to y is denoted by f_y or f / y and is obtained by treating x as a constant and differentiating f with respect to y .
- The partial derivatives of a function can be used to find the slope of a tangent line to a surface, the direction and rate of change of a function, and the optimization of a function .
- The partial derivatives of a function can be calculated using the same rules and formulas as ordinary derivatives, such as the power rule, the product rule, the quotient rule, and the chain rule .

Examples

- Find the partial derivatives of $f(x,y) = xy + x^2y$
 - $f_x = y + 2xy$ (by treating y as a constant and applying the power rule and the product rule)
 - $f_y = x + x^2$ (by treating x as a constant and applying the power rule and the product rule)
- Find the partial derivatives of $g(x,y) = \sin(x)\cos(y)$
 - $g_x = \cos(x)\cos(y)$ (by treating y as a constant and applying the chain rule)
 - $g_y = -\sin(x)\sin(y)$ (by treating x as a constant and applying the chain rule)
- Find the partial derivatives of $h(x,y) = e^{(x+y)/(x-y)}$
 - $h_x = (e^{(x+y)/(x-y)} + e^{(x+y)/(x-y)})/2$ (by treating y as a constant and applying the quotient rule and the chain rule)
 - $h_y = (e^{(x+y)/(x-y)} - e^{(x+y)/(x-y)})/2$ (by treating x as a constant and applying the quotient rule and the chain rule)

Euler's Theorem for Homogeneous Functions

- A function $f(x, y, z, \dots)$ is said to be **homogeneous** of degree n if it satisfies the condition

$$f(tx, ty, tz, \dots) = t^n f(x, y, z, \dots)$$

for any positive constant t and any values of $x, y, z, \dots$

- For example, the function $f(x, y, z) = x^3 + y^3 + z^3 + 3xyz$ is homogeneous of degree 3 because

$$f(tx, ty, tz) = (tx)^3 + (ty)^3 + (tz)^3 + 3(tx)(ty)(tz) = t^3(x^3 + y^3 + z^3 + 3xyz) = t^3 f(x, y, z).$$

- Euler's theorem** for homogeneous functions states that if $f(x, y, z, \dots)$ is a homogeneous function of degree n of k variables $x, y, z, \dots$, then

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} + \dots = nf.$$

This can be written in a compact form using the summation notation as

$$\sum_{i=1}^k x_i \frac{\partial f}{\partial x_i} = nf.$$

- To prove Euler's theorem, we can use the chain rule and the definition of homogeneous function. Let t be a positive constant and consider the function

$$g(t) = f(tx, ty, tz, \dots).$$

Then by the chain rule, we have

$$g'(t) = \frac{d}{dt}f(tx, ty, tz, \dots) = \sum_{i=1}^k \frac{\partial f}{\partial x_i} \frac{d}{dt}(tx_i) = \sum_{i=1}^k x_i \frac{\partial f}{\partial x_i} t.$$

On the other hand, by the definition of homogeneous function, we have

$$g(t) = f(tx, ty, tz, \dots) = t^n f(x, y, z, \dots).$$

Therefore,

$$g'(t) = \frac{d}{dt}(t^n f(x, y, z, \dots)) = nt^{n-1} f(x, y, z, \dots).$$

Equating the two expressions for $g'(t)$ and setting $t = 1$, we get

$$\sum_{i=1}^k x_i \frac{\partial f}{\partial x_i} = nf,$$

which is Euler's theorem.

Total derivative

- The total derivative of a function of several variables means the total change in the dependent variable due to the changes in all the independent variables.
- The total derivative is the derivative with respect to a variable that depends on the function not only directly but also via the intermediate variables.
- The formula for a total derivative is a direct result of the chain rule.
- The total derivative can be used to approximate the change in a function given small changes in the variables, or to analyze the sensitivity or error propagation of a function.

Definition and formula

- Suppose $z = f(x, y)$ be a function of two variables, where z is the dependent variable and x and y are the independent variables.
- The total derivative of z with respect to t is denoted by dz/dt and is defined as the limit of the ratio of the change in z to the change in t as the change in t approaches zero:

$$dz/dt = \lim_{\Delta t \rightarrow 0} (\Delta z)/(\Delta t)$$

- If x and y are also functions of t , then we can use the chain rule to express the total derivative as follows :

$$dz/dt = (dz/dx)(dx/dt) + (dz/dy)(dy/dt)$$

- The terms dz/dx and dz/dy are the partial derivatives of z with respect to x and y , respectively, and they measure the rate of change of z along the x and y directions.
- The terms dx/dt and dy/dt are the derivatives of x and y with respect to t , and they measure the rate of change of x and y along the t direction.

Example

- Suppose $z = x^2 + y^3$, where $x = t + 1$ and $y = 2t - 1$ are functions of t . Find the total derivative of z with respect to t .
- Solution:
 - We first find the partial derivatives of z with respect to x and y :

$$dz/dx = 2x$$

$$dz/dy = 3y^2$$
 - We then find the derivatives of x and y with respect to t :

$$dx/dt = 1$$

$$dy/dt = 2$$
 - We then use the formula for the total derivative to get:

$$dz/dt = (dz/dx)(dx/dt) + (dz/dy)(dy/dt)$$

$$= (2x)(1) + (3y^2)(2)$$

$$= 2x + 6y^2$$
 - We can also substitute $x = t + 1$ and $y = 2t - 1$ to get:

$$[dz/dt = 2(t + 1) + 6(2t - 1)^2](\text{https://latex.codecogs.com/png})$$

Change of variables for the notes of the Unit 2 - Differential Calculus- I in the subject of ENGINEERING MATHEMATICS-I

- Change of variables is a method of transforming a differential equation into a simpler or more convenient form by introducing a new dependent or independent variable.
- Change of variables can be useful for solving separable, homogeneous, or Bernoulli differential equations, as well as for evaluating integrals by substitution.
- Some examples of change of variables are:

- Substituting $u = y'$ to reduce a homogeneous differential equation of the form $y' = Q(x) - P(x)y$ to a separable equation of the form $u = Q(x) - P(x)u$.
- Substituting $y = xV(x)$ to reduce a homogeneous differential equation of the form $y' = f(x, y)$ to a separable equation of the form $\frac{1}{F(V)-V} \frac{dV}{dx} = \frac{1}{x}$.
- Substituting $y = vx^n$ to reduce a Bernoulli differential equation of the form $y' + p(x)y = q(x)y^n$ to a linear equation of the form $v' + (1-n)p(x)v = (1-n)q(x)$.
- Substituting $u = g(x)$ to evaluate an integral of the form $\int f(g(x))g'(x)dx$ by using the formula $\int f(g(x))g'(x)dx = \int f(u)du$.
- Change of variables can also be applied to multivariable functions, such as in the change of variables theorem for multiple integrals, which states that if T is a one-to-one and continuously differentiable transformation from a region D in the uv -plane to a region R in the xy -plane, then $\iint_R f(x, y)dA = \iint_D f(T(u, v))|J|dudv$, where J is the Jacobian determinant of the transformation.

Unit 3 - Differential Calculus-II

- This unit covers the following topics:
 - Applications of derivatives: optimization problems, related rates, curve sketching, linear approximation and differentials, mean value theorem, L'Hospital's rule, Newton's method.
 - Indeterminate forms and improper integrals: types of indeterminate forms, evaluation of limits using L'Hospital's rule, definition and properties of improper integrals, comparison test, convergence and divergence of improper integrals.
 - Infinite series: definition and examples of sequences and series, convergence and divergence tests, absolute and conditional convergence, power series, radius and interval of convergence, Taylor and Maclaurin series, applications of power series.
- The main objectives of this unit are:
 - To learn how to use derivatives to solve various problems involving optimization, rates of change, graphing, approximation, and numerical methods.
 - To understand the concept of indeterminate forms and how to apply L'Hospital's rule to evaluate limits of such forms.
 - To learn the concept of improper integrals and how to determine their convergence or divergence using comparison test and other methods.
 - To learn the concept of infinite series and how to test their convergence or divergence using various criteria.
 - To learn how to represent functions as power series and how to find the radius and interval of convergence of such series.

- To learn how to find the Taylor and Maclaurin series of a function and how to use them to approximate functions and evaluate limits.
- To learn how to use power series to solve differential equations and other applications.

Expansion of functions by Taylor's and Maclaurin's theorems for functions of one and two variables

- Taylor's theorem states that any infinitely differentiable function $f(x)$ can be approximated by a polynomial of degree n , called the Taylor polynomial, near a point a , such that the error term (or remainder) goes to zero as n goes to infinity.
- The Taylor polynomial of degree n for $f(x)$ at a is given by:

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

- The Taylor series of $f(x)$ at a is the infinite sum of the Taylor polynomials:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

- The Taylor series may or may not converge to the original function $f(x)$, depending on the function and the point a . A proof is required to show that they are equal (or not equal) for a function under consideration.
- The Maclaurin series is a special case of the Taylor series, where the point a is zero. The Maclaurin series of $f(x)$ at zero is given by:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

- The Maclaurin series is useful for expanding functions that are symmetric or periodic around the origin, such as trigonometric, exponential, and logarithmic functions.
- Taylor's theorem can be generalized to functions of two or more variables, such as $f(x,y)$. The Taylor polynomial of degree n for $f(x,y)$ at (a,b) is given by:

$$P_n(x,y) = \sum_{i=0}^n \sum_{j=0}^{n-i} \frac{1}{i!j!} \frac{\partial^{i+j} f}{\partial x^i \partial y^j}(a,b)(x-a)^i(y-b)^j$$

- The Taylor series of $f(x,y)$ at (a,b) is the infinite sum of the Taylor polynomials:

$$f(x, y) = \sum_{n=0}^{\infty} \sum_{i=0}^n \sum_{j=0}^{n-i} \frac{1}{i!j!} \frac{\partial^{i+j} f}{\partial x^i \partial y^j}(a, b)(x-a)^i(y-b)^j$$

- The Taylor series may or may not converge to the original function $f(x, y)$, depending on the function and the point (a, b) . A proof is required to show that they are equal (or not equal) for a function under consideration.
- The Maclaurin series for $f(x, y)$ at $(0, 0)$ is a special case of the Taylor series, where the point (a, b) is zero. The Maclaurin series of $f(x, y)$ at zero is given by:

$$f(x, y) = \sum_{n=0}^{\infty} \sum_{i=0}^n \sum_{j=0}^{n-i} \frac{1}{i!j!} \frac{\partial^{i+j} f}{\partial x^i \partial y^j}(0, 0)x^i y^j$$

- The Maclaurin series is useful for expanding functions that are symmetric or periodic around the origin, such as trigonometric, exponential, and logarithmic functions of two variables.

Maxima and Minima of Functions of Several Variables

- A function $f(x, y)$ of two independent variables has a **maximum** at a point (x_0, y_0) if $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) in the neighborhood of (x_0, y_0) . Such a function has a **minimum** at a point (x_0, y_0) if $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) in the neighborhood of (x_0, y_0) .
- The maximum and minimum values of a function are also called the **extrema** of the function.
- The highest and lowest values of a function within a particular set of ranges are known as **local maxima** and **local minima**. The highest and lowest values of the function under the whole range are known as the **absolute maxima** and the **absolute minima**.
- To find the local maxima and minima of a function $f(x, y)$, we need to find the points where the partial derivatives of $f(x, y)$ are zero or undefined. These points are called the **critical points** of the function.
- To find the absolute maxima and minima of a function $f(x, y)$ on a closed and bounded set D , we need to compare the values of $f(x, y)$ at the critical points inside D and at the boundary points of D .
- To classify the critical points of a function $f(x, y)$ as local maxima, local minima, or **saddle points**, we can use the **second derivative test**. The second derivative test involves finding the **Hessian matrix** of $f(x, y)$ and computing its **determinant** and **trace**.
- A saddle point is a point where the function has a local minimum in one direction and a local maximum in another direction. The graph of the function at a saddle point looks like a saddle.

- Some applications of finding the maxima and minima of functions of several variables include optimizing the area, volume, surface area, profit, cost, etc. of various objects or situations.

Lagrange's method of multipliers

- Lagrange's method of multipliers is a technique for finding the local maxima and minima of a function of several variables subject to one or more equality constraints .
- The basic idea is to construct a new function, called the Lagrangian, that combines the original function and the constraint function using a constant, called the Lagrange multiplier .
- The Lagrangian is defined as:

$$L(x, y, z, \lambda) = f(x, y, z) - \lambda(g(x, y, z) - k)$$

where $f(x, y, z)$ is the original function, $g(x, y, z) = k$ is the constraint, and λ is the Lagrange multiplier .

- The method of Lagrange multipliers states that the local extrema of $f(x, y, z)$ subject to $g(x, y, z) = k$ occur at the points where the gradient of the Lagrangian is zero, i.e.,

$$\nabla L(x, y, z, \lambda) = 0$$

This means that the partial derivatives of L with respect to x , y , z , and λ are all zero .

- The above equation implies that the gradient of f is parallel to the gradient of g , i.e.,

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$$

This means that the level curves of f and g are tangent at the points of extrema .

- To find the local extrema of $f(x, y, z)$ subject to $g(x, y, z) = k$, the following steps are followed :
 - Step 1: Construct the Lagrangian function $L(x, y, z, \lambda)$ as shown above.
 - Step 2: Solve the system of equations $\nabla L(x, y, z, \lambda) = 0$ for x , y , z , and λ .
 - Step 3: Plug in the solutions from Step 2 into $f(x, y, z)$ and compare the values to identify the maximum and minimum values, if they exist.

- Example: Find the maximum and minimum values of $f(x, y) = x^2 + y^2$ subject to the constraint $g(x, y) = x + y - 1 = 0$.

– Step 1: The Lagrangian function is

$$L(x, y, \lambda) = x^2 + y^2 - \lambda(x + y - 1)$$

– Step 2: The system of equations $\nabla L(x, y, \lambda) = 0$ is

$$\begin{aligned}\frac{\partial L}{\partial x} &= 2x - \lambda = 0 \\ \frac{\partial L}{\partial y} &= 2y - \lambda = 0 \\ \frac{\partial L}{\partial \lambda} &= x + y - 1 = 0\end{aligned}$$

Solving for x , y , and λ , we get

$$x = y = \frac{1}{2}, \lambda = 1$$

– Step 3: Plugging in the solution into $f(x, y)$, we get

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$$

This is the only possible value of f subject to the constraint, so it is both the maximum and minimum value. The point $(1/2, 1/2)$ is the only point of extrema.

Jacobians

- A **Jacobian matrix** is a matrix that contains the **first-order partial derivatives** of a **vector-valued function** of several variables .
- A vector-valued function is a function that takes one or more variables as input and outputs a vector, such as $\mathbf{f}(x, y) = (x^2 + y, y^2 - x)$.
- The Jacobian matrix can be used to **approximate the change** of the vector-valued function near a given point, by using a **linear transformation** .
- The Jacobian matrix can also be used to **convert integrals** from one coordinate system to another, by using the **Jacobian determinant** .
- The **Jacobian determinant** is the **determinant** of the Jacobian matrix, and it represents the **ratio of the area** (or volume) elements in the two coordinate systems .
- The Jacobian matrix and determinant depend on the **order** and **choice** of the variables in the vector-valued function and the coordinate system .

Example

- Consider the vector-valued function $\mathbf{f}(x, y) = (x^2 + y, y^2 - x)$ and the point $(1, 1)$.
- The Jacobian matrix of $\mathbf{f}$ at $(1, 1)$ is given by

$$\mathbf{J}(\mathbf{f})|_{(1,1)} = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} |_{(1,1)} = \begin{bmatrix} 2x & 1 \\ -1 & 2y \end{bmatrix} |_{(1,1)} = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

- The Jacobian determinant of $\mathbf{f}$ at $(1, 1)$ is given by

$$\det(\mathbf{J}(\mathbf{f}))|_{(1,1)} = 2 \times 2 - 1 \times (-1) = 5$$

- The Jacobian matrix can be used to approximate the change of $\mathbf{f}$ near $(1, 1)$, by using the formula

$$\mathbf{f}(x, y) \approx \mathbf{f}(1, 1) + \mathbf{J}(\mathbf{f})|_{(1,1)} \begin{bmatrix} x - 1 \\ y - 1 \end{bmatrix}$$

- The Jacobian determinant can be used to convert integrals from the Cartesian coordinate system (x, y) to the polar coordinate system (r, θ) , by using the formula

$$\int \int_R f(x, y) dx dy = \int \int_S f(r \cos \theta, r \sin \theta) |\det(\mathbf{J}(r, \theta))| dr d\theta$$

where R is the region in the Cartesian plane, S is the region in the polar plane, and $\mathbf{J}(r, \theta)$ is the Jacobian matrix of the transformation from (x, y) to (r, θ) , which is given by

$$\mathbf{J}(r, \theta) = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$$

and $\det(\mathbf{J}(r, \theta)) = r$.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic of approximation of errors for the notes of the Unit 3 - Differential Calculus-II in the subject of ENGINEERING MATHEMATICS-I.

Approximation of errors

- In many practical situations, we need to estimate the error or uncertainty in a quantity that is calculated from other quantities with known errors.

- For example, if we measure the length and width of a rectangle with a ruler, we can calculate its area, but we also need to know how accurate our calculation is, given the possible errors in the measurements.
- One way to estimate the error in a function of several variables is to use the **differential approximation**. This method is based on the idea that if the variables change by small amounts, then the function changes by approximately the linear combination of the partial derivatives times the changes in the variables.
- Mathematically, we can write this as:

$$f(x + \Delta x, y + \Delta y) \approx f(x, y) + f_x(x, y)\Delta x + f_y(x, y)\Delta y$$

- where Δx and Δy are the errors or uncertainties in x and y , respectively, and f_x and f_y are the partial derivatives of f with respect to x and y , respectively, evaluated at the point (x, y) .
- The difference between the left-hand side and the right-hand side of the equation is called the **error in the differential approximation**, and it represents the amount by which the linear approximation deviates from the actual value of the function.
- We can use the differential approximation to estimate the error in the function value as follows:

$$\text{Error in } f \approx f_x(x, y)\Delta x + f_y(x, y)\Delta y$$

- Note that this is only an approximation, and it becomes more accurate as the errors in the variables become smaller. Also, this method does not account for the possible correlations or dependencies between the variables, which may affect the error propagation.
- To find the **maximum possible error** in the function value, we need to use the **absolute values** of the partial derivatives and the errors, and add them up. This gives us the following formula:

$$\text{Maximum error in } f \leq |f_x(x, y)|\Delta x + |f_y(x, y)|\Delta y$$

- This is also known as the **first-order approximation** of the error, or the **total differential** of the function. It gives us an upper bound for the error, but it does not tell us the exact value or the direction of the error.
- To find the **relative error** or the **percentage error** in the function value, we need to divide the error by the function value and multiply by 100%. This gives us the following formulas:

$$\text{Relative error in } f \approx \frac{f_x(x, y)\Delta x + f_y(x, y)\Delta y}{f(x, y)}$$

$$\text{Percentage error in } f \approx \frac{f_x(x, y)\Delta x + f_y(x, y)\Delta y}{f(x, y)} \times 100\%$$

- These formulas give us an estimate of how large the error is compared to the function value, and they are useful for comparing the errors in different functions or situations.
- To illustrate the use of these formulas, let us consider an example.

Example

- Suppose we want to calculate the volume of a cylindrical tank with radius r and height h , given by the formula $V = \pi r^2 h$. We measure the radius and the height with a tape measure, and we find that $r = 2.5 \pm 0.1$ m and $h = 4 \pm 0.2$ m, where the $\pm$ signs indicate the possible errors in the measurements. How can we estimate the error in the volume?
- Solution:
 - To use the differential approximation, we need to find the partial derivatives of the volume function with respect to the radius and the height. We have:

$$V_r = \frac{\partial V}{\partial r} = 2\pi r h$$

$$V_h = \frac{\partial V}{\partial h} = \pi r^2$$

- Evaluating these at the point $(r, h) = (2.5, 4)$, we get:

$$V_r(2.5, 4) = 2\pi(2.5)(4) = 20\pi$$

Unit 4 - Multiple integration

- Multiple integration is the extension of single-variable integration to functions of two or more variables, such as $f(x, y)$ or $f(x, y, z)$.
- Multiple integration can be used to calculate areas, volumes, masses, centroids, moments of inertia, and other geometric or physical quantities of regions in the plane or in space.
- Multiple integration can also be used to evaluate integrals that cannot be solved by single-variable methods, such as integrals involving trigonometric functions, exponential functions, or logarithmic functions of several variables.
- The main types of multiple integrals are:
 - Double integrals: These are integrals of functions of two variables over a region in the xy -plane. They can be computed by iterated integration, where the integral is first evaluated with respect to one variable, and then with respect to the other variable. The order of integration can be changed if the region and the function allow it. Double integrals can also be computed by changing to polar coordinates, which can simplify the region and the integrand.

- Triple integrals: These are integrals of functions of three variables over a region in the xyz-space. They can be computed by iterated integration, where the integral is first evaluated with respect to one variable, then with respect to another variable, and finally with respect to the third variable. The order of integration can be changed if the region and the function allow it. Triple integrals can also be computed by changing to cylindrical or spherical coordinates, which can simplify the region and the integrand.
- Line integrals: These are integrals of functions of two or three variables along a curve in the plane or in space. They can be computed by parametrizing the curve, and then integrating the function along the parameter interval. Line integrals can be used to calculate the work done by a force field along a path, or the circulation of a vector field around a closed curve.
- Surface integrals: These are integrals of functions of three variables over a surface in space. They can be computed by parametrizing the surface, and then integrating the function over the parameter domain. Surface integrals can be used to calculate the flux of a vector field across a surface, or the area of a surface.
- Multiple integration can be generalized to higher dimensions, where integrals of functions of more than three variables are defined over regions in higher-dimensional spaces. These integrals can be computed by iterated integration, or by changing to appropriate coordinate systems.

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Double integral

- A double integral is a way to integrate over a two-dimensional area. It can be used to find the volume under a surface, the area of a region, the average value of a function, and other applications.
- A double integral of a function of two variables, say $f(x,y)$, over a region R in the xy -plane, is denoted by:

$$\iint_R f(x,y) dA$$

where dA is a small element of area in the region R .

- A double integral can be evaluated by iterated integration, which means integrating first with respect to one variable, then with respect to the other variable. For example, if R is a rectangle with vertices (a,b) , (a,d) , (c,b) , and (c,d) , then the double integral can be written as:

$$\iint_R f(x, y) dA = \int_a^c \int_b^d f(x, y) dy dx = \int_b^d \int_a^c f(x, y) dx dy$$

- The order of integration can be changed if the region R is simple enough, and the function $f(x, y)$ is continuous. The order of integration can affect the ease of computation and the appearance of the integral.
- A double integral can also be evaluated by changing to polar coordinates, which are more suitable for regions that are circular or symmetric. In polar coordinates, a point (x, y) is represented by (r, θ) , where r is the distance from the origin and θ is the angle from the positive x -axis. The element of area dA is given by:

$$dA = r dr d\theta$$

- The region R in polar coordinates is determined by the limits of r and θ . For example, if R is a circle with radius a centered at the origin, then the double integral can be written as:

$$\iint_R f(x, y) dA = \int_0^{2\pi} \int_0^a f(r \cos \theta, r \sin \theta) r dr d\theta$$

- A double integral can also be evaluated by changing to other coordinate systems, such as cylindrical or spherical coordinates, depending on the shape of the region and the function.
- A double integral can be used to calculate the volume of a solid bounded by a surface $z = f(x, y)$ and the xy -plane, by setting the function to be 1. For example, if the solid is a cylinder with radius a and height h , then the double integral can be written as:

$$\iint_R 1 dA = \int_0^{2\pi} \int_0^a 1 r dr d\theta = \pi a^2 h$$

- A double integral can be used to calculate the area of a region R in the xy -plane, by setting the function to be 1. For example, if the region is a triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$, then the double integral can be written as:

$$\iint_R 1 dA = \int_0^1 \int_0^x 1 dy dx = \frac{1}{2}$$

- A double integral can be used to calculate the average value of a function $f(x, y)$ over a region R in the xy -plane, by dividing the integral by the area of the region. For example, if the region is a square with side length 2 and the function is $f(x, y) = x^2 + y^2$, then the average value is:

$$\frac{\iint_R f(x, y) dA}{\iint_R 1 dA} = \frac{\int_{-1}^1 \int_{-1}^1 (x^2 + y^2) dy dx}{\int_{-1}^1 \int_{-1}^1 1 dy dx} = \frac{8}{3}$$

- A double integral can be used to calculate other quantities, such as the center of mass, the moment of inertia, the work done by a force, and the flux of a vector field, by choosing appropriate functions and

Triple integral

- A triple integral is an iterated integral with three variables and over a three-dimensional region.
- A triple integral can be used to calculate the volume, mass, center of mass, moment of inertia, and other properties of a solid region.
- A triple integral can be written as $\iiint_R f(x, y, z) dV$, where $f(x, y, z)$ is a function of three variables and dV is a differential volume element.
- A triple integral can be evaluated by integrating $f(x, y, z)$ with respect to one variable, then integrating the result with respect to another variable, and finally integrating the final result with respect to the third variable.
- The order of integration can be changed as long as the limits of integration are adjusted accordingly.
- A triple integral can be expressed in different coordinate systems, such as Cartesian, cylindrical, or spherical, depending on the shape and symmetry of the region of integration.
- A triple integral can be transformed by using a change of variables, such as $u = g(x, y, z)$, $v = h(x, y, z)$, $w = k(x, y, z)$, where g , h , and k are differentiable functions with nonzero Jacobian determinant.
- A triple integral can be used to compute the average value of a function over a solid region by dividing the integral by the volume of the region.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic of change of order of integration for the notes of the Unit 4 - Multiple integration in the subject of ENGINEERING MATHEMATICS-I.

Change of order of integration

- Sometimes, it is easier to evaluate a double integral by changing the order of integration.
- The order of integration is the order in which the integrals are performed.
- The order of integration can be changed by using the following steps:
 1. Sketch the region of integration and identify the limits of integration for each variable.

2. Rewrite the limits of integration by interchanging the roles of the variables and the functions that define the region.
 3. Evaluate the new double integral by performing the integrals in the reversed order.
- Example: Evaluate the double integral $\int_0^1 \int_x^{\sqrt{x}} f(x, y) dy dx$ by changing the order of integration.
 - Solution: The region of integration is bounded by the curves $y = x$, $y = \sqrt{x}$, and $x = 1$. The limits of integration for x are 0 and 1, and the limits of integration for y are x and $\sqrt{x}$.
 - To change the order of integration, we interchange the roles of x and y . The limits of integration for y are 0 and 1, and the limits of integration for x are y^2 and y . The new double integral is $\int_0^1 \int_{y^2}^y f(x, y) dx dy$.
 - To evaluate the new double integral, we perform the integrals in the reversed order. First, we integrate with respect to x , then with respect to y . The result is $\int_0^1 [F(y, y) - F(y^2, y)] dy$, where $F(x, y)$ is an antiderivative of $f(x, y)$ with respect to x .

Change of variables for the notes of the Unit 4 - Multiple integration in the subject of ENGI- NEERING MATHEMATICS-I

- The change of variables in multiple integrals is a technique that allows us to simplify the integration of a function over a complex region by transforming it to a function over a simpler region.
- The change of variables in multiple integrals is based on the idea of planar transformations, which are functions that map one region to another by changing their variables.
- For example, we can transform a region R in the xy -plane to a region R' in the uv -plane by using the functions $x = x(u, v)$ and $y = y(u, v)$.
- The inverse transformation is given by $u = u(x, y)$ and $v = v(x, y)$.
- The change of variables formula for multiple integrals states that if $x = x(u, v)$ and $y = y(u, v)$ define a one-to-one mapping of R' onto R , then

$$\iint_R f(x, y) dA = \iint_{R'} f(x(u, v), y(u, v)) |J(u, v)| du dv$$

where $J(u, v)$ is the Jacobian determinant of the transformation, given by

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

- The Jacobian determinant measures the local change of area induced by the transformation. It can be positive or negative depending on the orientation of the regions.
- The change of variables formula can be extended to higher dimensions by using more variables and larger Jacobian determinants.
- The change of variables technique can be useful for evaluating integrals that are difficult or impossible to do in the original variables, such as integrals involving polar, cylindrical, or spherical coordinates.

Beta and Gamma Function and Their Properties

Definition of Beta Function

The beta function, also known as the Euler integral of the first kind, is a function of two variables that is defined as

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt$$

for any positive real numbers x and y .

Definition of Gamma Function

The gamma function, also known as the Euler integral of the second kind, is a function of one variable that is defined as

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$$

for any positive real number x .

Relationship between Beta and Gamma Function

A key property of the beta function is its close relationship to the gamma function:

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

A proof of this formula can be found in or .

Properties of Beta Function

Some of the properties of the beta function are:

- The beta function is symmetric, meaning that $B(x, y) = B(y, x)$ for all x and y .
- The beta function is closely related to binomial coefficients, as $B(n+1, m+1) = \frac{n!m!}{(n+m+1)!}$ for any non-negative integers n and m .
- The beta function satisfies the recurrence relation $B(x+1, y) = \frac{x}{x+y}B(x, y+1)$ for all x and y .
- The beta function can be expressed in terms of the hypergeometric function as $B(x, y) = \frac{x}{y} {}_2F_1(1, x; x+y; 1)$ for all x and y .

Properties of Gamma Function

Some of the properties of the gamma function are:

- The gamma function is a generalization of the factorial function, as $\Gamma(n) = (n-1)!$ for any positive integer n .
- The gamma function satisfies the functional equation $\Gamma(x+1) = x\Gamma(x)$ for all x .
- The gamma function has a unique analytic continuation to the complex plane, except for the negative integers, where it has simple poles.
- The gamma function can be expressed in terms of the incomplete gamma function as $\Gamma(x) = \gamma(x, 0) + \Gamma(x, 0)$ for all x , where $\gamma(x, a) = \int_0^a t^{x-1}e^{-t}dt$ and $\Gamma(x, a) = \int_a^\infty t^{x-1}e^{-t}dt$ are the lower and upper incomplete gamma functions, respectively.
- The gamma function can be approximated by Stirling's formula as $\Gamma(x) \approx \sqrt{2\pi x} \left(\frac{x}{e}\right)^x$ for large x .

Dirichlet's integral and its applications to area and volume

- Dirichlet's integral is a type of integral that appears in various contexts in mathematics and physics, such as Dirichlet's principle, Fourier series, and phase space.
- One form of Dirichlet's integral is given by

$$D(f) = \int_{\Omega} |\nabla f|^2 d\Omega$$

where Ω is a domain in $\mathbb{R}^n$, f is a function defined on Ω , and ∇f is the gradient of f . - Dirichlet's principle states that a function f that minimizes the Dirichlet integral $D(f)$ among all functions that satisfy a given boundary condition is a solution to the Laplace equation $\Delta f = 0$ on Ω . - Dirichlet's integral can also be used to evaluate the phase volume of a physical system, which is the volume of

the region in phase space occupied by the system. Phase space is the space of all possible states of the system, such as position and momentum. For example, for a particle moving in one dimension, the phase space is the (x, p) plane, where x is the position and p is the momentum of the particle. - Dirichlet's integral formula for the phase volume is given by

$$V = \frac{1}{(2\pi\hbar)^n} \int_{\Omega} \exp\left(-\frac{i}{\hbar} S(x, p)\right) dx dp$$

where n is the number of degrees of freedom of the system, $\hbar$ is the reduced Planck constant, Ω is the region in phase space, and $S(x, p)$ is the action function of the system, which is the integral of the Lagrangian along a path in phase space. - Dirichlet's integral can also be applied to find the area and volume of surfaces and solids in $\mathbb{R}^3$. For example, if $x(u, v)$ is a parametric representation of a surface in $\mathbb{R}^3$, then the area of the surface is given by

$$A = \int_B |x_u \times x_v| du dv$$

where B is the domain of the parameters (u, v) , x_u and x_v are the partial derivatives of x with respect to u and v , and $\times$ is the cross product. Similarly, if $x(u, v, w)$ is a parametric representation of a solid in $\mathbb{R}^3$, then the volume of the solid is given by

$$V = \int_B |x_u \cdot (x_v \times x_w)| du dv dw$$

where B is the domain of the parameters (u, v, w) , x_u , x_v , and x_w are the partial derivatives of x with respect to u , v , and w , and $\cdot$ is the dot product.

Liouville's extensions of Dirichlet's integral

- Dirichlet's integral is a special case of a multiple integral of the form

$$\int_0^\infty \int_0^\infty \frac{x^{a-1} y^{b-1}}{(x+y)^{a+b}} dx dy = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

where Γ is the gamma function and $a, b > 0$.

- Liouville's extension of Dirichlet's integral is a generalization that allows the integration region to be any bounded domain V in $\mathbb{R}^n$ and the integrand to be any function $F(x_1, x_2, \dots, x_n)$ that depends only on the sum of the variables $x_1 + x_2 + \dots + x_n = h$.

- Liouville's extension states that if $x_1, x_2, \dots, x_n$ are all positive such that $h_1 < h < h_2$, then

$$\int_V x_1^{a_1-1} x_2^{a_2-1} \dots x_n^{a_n-1} F(x_1 + x_2 + \dots + x_n) dx_1 dx_2 \dots dx_n = \frac{\Gamma(a_1) \Gamma(a_2) \dots \Gamma(a_n)}{\Gamma(a_1 + a_2 + \dots + a_n)} \int_{h_1}^{h_2} F(h) h^{a_1+a_2+\dots+a_n-1} dh$$

where $a_1, a_2, \dots, a_n > 0$.

- The proof of Liouville's extension is based on the change of variables $x_1 + x_2 + \dots + x_n = h$ and $x_1/h = u_1, x_2/h = u_2, \dots, x_n/h = u_n$, which transforms the multiple integral into a product of a single integral and a beta function.
- Liouville's extension can be used to evaluate multiple integrals that involve functions of the sum of the variables, such as

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{1}{(1+x+y+z)^4} dz dy dx = \frac{\pi^2}{36}$$

by applying the extension with $n = 4$, $a_1 = a_2 = a_3 = a_4 = 1$, $F(h) = 1/(1+h)^4$, $h_1 = 0$, and $h_2 = 1$.

Unit 5 - Vector Calculus

Vector calculus is a branch of mathematics that deals with the differentiation and integration of vector fields, primarily in 3-dimensional Euclidean space. Vector fields represent the distribution of a given vector to each point in the subset of the space. Vectors are quantities that have both magnitude and direction, and can be represented by arrows in a coordinate system.

Some of the topics covered in vector calculus are:

- Vector operations: such as addition, subtraction, scalar multiplication, dot product, cross product, and norm. These operations have both algebraic and geometric interpretations, and can be used to find the length, angle, area, volume, and projection of vectors .
- Vector functions: such as parametric curves, arc length, curvature, and torsion. These functions describe the motion of a particle or a curve in space, and can be used to find the speed, acceleration, and tangential and normal components of the motion.
- Vector fields: such as gradient, divergence, curl, and Laplacian. These fields measure the rate of change of a scalar or vector function in different directions, and can be used to find the directional derivative, the normal vector, the potential function, and the divergence and curl theorems.

- Line integrals: such as work, circulation, and flux. These integrals evaluate the integral of a scalar or vector function along a curve, and can be used to find the work done by a force, the circulation of a vector field, and the flux of a vector field across a curve.
- Surface integrals: such as area, flux, and divergence theorem. These integrals evaluate the integral of a scalar or vector function over a surface, and can be used to find the area of a surface, the flux of a vector field across a surface, and the divergence theorem, which relates the flux of a vector field across a closed surface to the divergence of the vector field inside the surface.
- Volume integrals: such as volume, mass, and moment of inertia. These integrals evaluate the integral of a scalar or vector function over a volume, and can be used to find the volume of a solid, the mass of a solid with density function, and the moment of inertia of a solid with respect to an axis.
- Green's theorem: which relates the line integral of a vector field along a closed curve to the double integral of the curl of the vector field over the region enclosed by the curve.
- Stokes' theorem: which relates the line integral of a vector field along a closed curve to the surface integral of the curl of the vector field over the surface bounded by the curve.

Vector calculus is particularly useful in studying physical phenomena such as center of mass, field theory, kinematics, and Maxwell's equations.

Vector differentiation: Gradient

- The gradient of a scalar-valued function of several variables is a vector-valued function that measures the direction and rate of fastest increase of the function at a given point.
- The gradient is denoted by the symbol ∇ (nabla) and is defined as the vector of partial derivatives of the function with respect to each variable.
- For example, if $f(x,y,z)$ is a scalar-valued function of three variables, then the gradient of f is given by

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

where $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are the unit vectors along the x , y , and z axes, respectively. - The gradient of f at a point (x_0, y_0, z_0) is a vector that points in the direction of the steepest ascent of f at that point. The magnitude of the gradient is equal to the slope of the tangent plane to the level surface of f at that point. - The gradient has several properties and applications in vector calculus, such as:

- The gradient is perpendicular to the level curves or surfaces of the function.

- The gradient is invariant under rotations and translations of the coordinate system.
- The gradient can be used to find the directional derivative of the function along any direction.
- The gradient can be used to find the normal vector to a surface defined by an implicit equation.
- The gradient can be used to find the equation of a tangent plane or a normal line to a surface at a given point.
- The gradient can be used to find the maximum and minimum values of a function subject to a constraint using the method of Lagrange multipliers.
- The gradient can be used to find the divergence and curl of a vector field, which are important concepts in fluid mechanics and electromagnetism.

Curl and Divergence and their Physical interpretation

- Curl and divergence are two operators that can be applied to vector fields in three-dimensional Euclidean space.
- Vector fields can be used to model various physical phenomena, such as fluid flow, electric and magnetic fields, heat and mass transfer, etc.
- Curl and divergence measure different aspects of the behavior of a vector field at a point.

Curl

- The curl of a vector field $\vec{F}$, denoted by $\nabla \times \vec{F}$, is a vector field that points in the direction of the axis of rotation of the fluid around that point, and has a magnitude equal to the angular speed of the rotation.
- The curl can be computed using the following formula, where i , j , and k are the unit vectors along the x , y , and z axes, respectively, and $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$, and $\frac{\partial}{\partial z}$ are the partial derivatives with respect to x , y , and z , respectively:

$$\nabla \times \vec{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) i + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) j + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) k$$

- The curl can also be computed using the determinant of the following matrix, where ∇ is the vector differential operator:

$$\nabla \times \vec{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

- The physical interpretation of the curl is the tendency of the fluid to swirl or rotate around a point. For example, if the curl of a vector field is zero at a point, then the fluid does not rotate around that point. If the curl is

nonzero, then the fluid rotates around that point, and the direction and magnitude of the curl indicate the axis and speed of the rotation.

Divergence

- The divergence of a vector field $\vec{F}$, denoted by $\nabla \cdot \vec{F}$, is a scalar field that measures the net rate of flow of the fluid out of a small region around that point.
- The divergence can be computed using the following formula, where $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$, and $\frac{\partial}{\partial z}$ are the partial derivatives with respect to x , y , and z , respectively:

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

- The divergence can also be computed using the dot product of the vector differential operator ∇ and the vector field $\vec{F}$:

$$\nabla \cdot \vec{F} = \nabla \bullet \vec{F}$$

- The physical interpretation of the divergence is the tendency of the fluid to collect or disperse at a point. For example, if the divergence of a vector field is zero at a point, then the fluid is neither created nor destroyed at that point, and the flow is incompressible. If the divergence is positive, then the fluid is created or flows out of that point, and the flow is divergent. If the divergence is negative, then the fluid is destroyed or flows into that point, and the flow is convergent.

Directional derivatives

- A directional derivative is a measure of the rate of change of a function in a given direction at a given point.
- It generalizes the concept of partial derivatives, which are the rates of change of a function in the x , y and z directions.
- The directional derivative of a function $f(x,y,z)$ at a point P in the direction of a unit vector $\mathbf{v}$ is denoted by $D_{\mathbf{v}}f(P)$ or $\mathbf{v}f(P)$ and is defined as:

$$D_{\mathbf{v}}f(P) = \mathbf{v}f(P) = \lim_{h \rightarrow 0} [f(P+h\mathbf{v}) - f(P)]/h$$

- Alternatively, the directional derivative can be expressed using the gradient of f , which is a vector that points in the direction of the greatest increase of f and has a magnitude equal to the rate of change in that direction. The gradient of f is denoted by ∇f or $\text{grad } f$ and is defined as:

$$\nabla f = \text{grad } f = \left(\frac{\partial f}{\partial x} \right) \mathbf{i} + \left(\frac{\partial f}{\partial y} \right) \mathbf{j} + \left(\frac{\partial f}{\partial z} \right) \mathbf{k}$$

- The directional derivative can then be computed as the dot product of the gradient and the unit vector $\mathbf{v}$:

$$D_{\mathbf{v}}f(\mathbf{P}) = \nabla f(\mathbf{P}) \cdot \mathbf{v}$$

- The directional derivative has the following properties:
 - It is a scalar quantity, not a vector.
 - It is zero if $\mathbf{v}$ is perpendicular to the gradient of f at $\mathbf{P}$, meaning that f does not change in that direction.
 - It is positive if $\mathbf{v}$ has an acute angle with the gradient of f at $\mathbf{P}$, meaning that f increases in that direction.
 - It is negative if $\mathbf{v}$ has an obtuse angle with the gradient of f at $\mathbf{P}$, meaning that f decreases in that direction.
 - It is equal to the magnitude of the gradient of f at $\mathbf{P}$ if $\mathbf{v}$ is parallel to the gradient of f at $\mathbf{P}$, meaning that f changes at the maximum rate in that direction.
- Example: Find the directional derivative of the function $f(x,y) = x^2 + y^2$ at the point $(1,1)$ in the direction of the vector $\mathbf{v} = 2\mathbf{i} - \mathbf{j}$.

- Solution: First, we need to find the gradient of f :

$$\nabla f = (f_x)\mathbf{i} + (f_y)\mathbf{j} = (2x)\mathbf{i} + (2y)\mathbf{j}$$

At the point $(1,1)$, the gradient is:

$$\nabla f(1,1) = (2)\mathbf{i} + (2)\mathbf{j}$$

- Next, we need to normalize the vector $\mathbf{v}$ to get a unit vector:

$$\mathbf{v} = 2\mathbf{i} - \mathbf{j}$$

$$|\mathbf{v}| = \sqrt{(2)^2 + (-1)^2} = \sqrt{5}$$

$$\mathbf{v}/|\mathbf{v}| = (2/\sqrt{5})\mathbf{i} - (1/\sqrt{5})\mathbf{j}$$

- Finally, we can compute the directional derivative as the dot product of the gradient and the unit vector:

$$\begin{aligned} D_{\mathbf{v}}f(1,1) &= \nabla f(1,1) \cdot \mathbf{v}/|\mathbf{v}| \\ &= [(2)\mathbf{i} + (2)\mathbf{j}] \cdot [(2/\sqrt{5})\mathbf{i} - (1/\sqrt{5})\mathbf{j}] \\ &= (4/\sqrt{5}) - (2/\sqrt{5}) \\ &= 2/\sqrt{5} \end{aligned}$$

$$0.8944$$

- This means that the function $f(x,y)$ increases at a rate of about 0.8944 units per unit distance in the direction of the vector $\mathbf{v}$ at the point $(1,1)$.

Vector Integration: Line integral

- A line integral is an integral in which a function is integrated along some curve in the coordinate system.
- The function which is to be integrated can either be represented as a scalar field or vector field. We can integrate both scalar-valued function and vector-valued function along a curve.
- A line integral of a scalar field is thus a line integral of a vector field, where the vectors are always tangential to the line of the integration.
- A line integral of a vector field can be thought of as a measure of the total effect of a given tensor field along a given curve. For example, the line integral over a scalar field can be interpreted as the area under the field carved out by a particular curve.
- Line integrals are useful in physics for computing the work done by a force on a moving object.
- The line integral of a vector field on a curve is defined by:

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

where $\mathbf{F}$ is the vector field, C is the curve, and $\cdot$ denotes a dot product. - In Cartesian coordinates, the line integral can be written as:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

where $\mathbf{r}(t)$ is a parametrization of the curve C with $a \leq t \leq b$ and $\mathbf{r}'(t)$ is the derivative of $\mathbf{r}(t)$ with respect to t . - We can also write line integrals of vector fields as a line integral with respect to arc length as follows:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

where $\mathbf{T}(t)$ is the unit tangent vector and is given by:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$$

and ds is the differential of arc length and is given by:

$$ds = \|\mathbf{r}'(t)\| dt$$

- The value of the line integral depends on the orientation of the curve C . If we parameterize the curve such that we move in the opposite direction as t increases, the value of the line integral is multiplied by -1 .

- The line integral of a vector field is independent of the parametrization $\mathbf{r}(t)$ in absolute value, but they do depend on its orientation.

Surface Integral

- A surface integral is a generalization of a line integral to account for surfaces in three dimensions.
- A surface integral is used to add a bunch of values associated with points on a surface, such as area, mass, flux, etc.
- A surface integral can be of two types: scalar or vector.
- A scalar surface integral is used to integrate a scalar function over a surface, such as the surface area of a sphere.
- A vector surface integral is used to integrate a vector field over a surface, such as the electric flux through a closed surface.
- A surface integral can be computed by using a parameterization of the surface, such as spherical or cylindrical coordinates, and applying the change of variables formula.
- A surface integral can also be computed by using the divergence theorem or the Stokes' theorem, which relate surface integrals to volume integrals or line integrals, respectively.

Examples

- Example 1: Find the surface area of a hemisphere of radius r .
 - Solution: The surface of a hemisphere can be parameterized by using spherical coordinates as follows:
 - * $x = r \sin \theta \cos \phi$
 - * $y = r \sin \theta \sin \phi$
 - * $z = r \cos \theta$
 where $0 \leq \theta \leq \pi/2$ and $0 \leq \phi \leq 2\pi$.
 - The surface element dS can be found by taking the cross product of the partial derivatives of the parameterization with respect to θ and ϕ :
 - * $dS = \left| \frac{\partial (x, y, z)}{\partial \theta} \times \frac{\partial (x, y, z)}{\partial \phi} \right| d\theta d\phi$
 - * $dS = r^2 \sin \theta d\theta d\phi$
 - The surface area of the hemisphere is then given by the scalar surface integral:
 - * $A = \int_S dS$
 - * $A = \int_0^{\pi/2} \int_0^{2\pi} r^2 \sin \theta d\phi d\theta$
 - * $A = r^2 \int_0^{\pi/2} [-\cos \theta]_0^{\pi/2} d\theta$
 - * $A = r^2 (1 - 0) (2\pi - 0)$

$$* A = 2r^2$$

- Example 2: Find the electric flux through a cube of side length a centered at the origin, if the electric field is given by $E = (x, y, z)$.

– Solution: The electric flux through a surface is given by the vector surface integral:

$$* \Phi = \int_S \mathbf{E} \cdot d\mathbf{S}$$

where $d\mathbf{S}$ is the outward normal vector to the surface.

– The cube has six faces, each of which can be parameterized by using Cartesian coordinates as follows:

- * Face 1: $x = a/2, -a/2 \leq y \leq a/2, -a/2 \leq z \leq a/2$
- * Face 2: $x = -a/2, -a/2 \leq y \leq a/2, -a/2 \leq z \leq a/2$
- * Face 3: $y = a/2, -a/2 \leq x \leq a/2, -a/2 \leq z \leq a/2$
- * Face 4: $y = -a/2, -a/2 \leq x \leq a/2, -a/2 \leq z \leq a/2$
- * Face 5: $z = a/2, -a/2 \leq x \leq a/2, -a/2 \leq y \leq a/2$
- * Face 6: $z = -a/2, -a/2 \leq x \leq a/2, -a/2 \leq y \leq a/2$

– The surface element $d\mathbf{S}$ for each face can be found by taking the cross product of the partial derivatives of the parameterization with respect to x and y or y and z or z and x , depending on the face, and multiplying by the appropriate sign to get the outward normal vector:

- * $d\mathbf{S}_1 = (a/2, 0, 0) dy dz$
- * $d\mathbf{S}_2 = (-a/2, 0, 0) dy dz$
- * $d\mathbf{S}_3 = (0, a/2, 0) dx dz$
- * $d\mathbf{S}_4 = (0, -a/2, 0) dx dz$

Volume integral

- A volume integral is a type of multiple integral that extends the concept of area integral to three-dimensional regions.
- A volume integral can be used to find the volume of a solid, the mass of a solid with varying density, the electric charge of a region with varying charge density, and other physical quantities that involve integration over a volume.
- A volume integral can be expressed in Cartesian, cylindrical, or spherical coordinates, depending on the shape and symmetry of the region of integration.
- A volume integral can be evaluated by using the following formula:

$$\iiint_V f(x, y, z) dV = \iiint_V f(x, y, z) dx dy dz$$

where $f(x, y, z)$ is the integrand function, V is the region of integration, and dV is the differential volume element.

- A volume integral can also be computed by using the divergence theorem or the Stokes' theorem, which relate the volume integral of a vector field to the surface integral of its divergence or curl, respectively. These theorems are useful for simplifying the calculation of volume integrals in some cases.

Gauss's Divergence Theorem

- Gauss's divergence theorem, also known as Gauss's theorem or Ostrogradsky's theorem, is a theorem in vector calculus that relates the flux of a vector field through a closed surface to the divergence of the field in the volume enclosed.
- The flux of a vector field is the amount of the field that passes through a given surface per unit time. The divergence of a vector field is a measure of how much the field diverges or spreads out from a given point.
- The theorem can be stated as follows: Let $\mathbf{V}$ be a region in space with boundary $\mathbf{S}$, and let $\mathbf{F}$ be a vector field that is continuously differentiable in $\mathbf{V}$. Then

$$\iint_{\mathbf{S}} \mathbf{F} \cdot d\mathbf{S} = \iiint_{\mathbf{V}} \nabla \cdot \mathbf{F} dV$$

- The left-hand side of the equation is the surface integral of $\mathbf{F}$ over $\mathbf{S}$, which represents the net flux of $\mathbf{F}$ out of the region $\mathbf{V}$. The right-hand side of the equation is the volume integral of the divergence of $\mathbf{F}$ over $\mathbf{V}$, which represents the net source or sink of $\mathbf{F}$ inside the region $\mathbf{V}$.
- The theorem can be interpreted as a conservation law: the net flux of $\mathbf{F}$ out of a region is equal to the net source or sink of $\mathbf{F}$ inside the region.
- The theorem can be proved using the divergence theorem in two dimensions, also known as Green's theorem, and applying it to each face of a small rectangular box that is contained in $\mathbf{V}$. By taking the limit as the box shrinks to a point, the theorem follows.
- The theorem can be generalized to higher dimensions and to more general surfaces and vector fields, as long as certain conditions are met. For example, the theorem holds for surfaces that are piecewise smooth and orientable, and for vector fields that are continuously differentiable and have compact support.
- The theorem has many applications in physics and engineering, such as calculating the electric flux through a closed surface using Gauss's law, or calculating the mass flow rate through a pipe using the continuity equation.

Green's theorem and Stoke's theorem (without proof) and their applications

Green's theorem

- Green's theorem is a formula that relates a line integral around a simple closed curve C to a double integral over the plane region D bounded by C .
- Green's theorem can be written as:

$$\oint_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

- where P and Q are functions of x and y that have continuous first partial derivatives on an open region that contains D .
- Green's theorem is a special case of the Kelvin–Stokes theorem, when applied to a region in the xy -plane.
- Green's theorem can be used to evaluate line integrals without parametrizing the curve C , by finding a related double integral over the region D .
- Green's theorem can also be used to find the area of a plane region D , by choosing P and Q such that their partial derivatives are equal to 1 and -1, respectively.
- Green's theorem can also be used to find the circulation and curl of a vector field along a curve C , by choosing P and Q to be the components of the vector field.

Stokes' theorem

- Stokes' theorem is a formula that relates a surface integral of the curl of a vector field over a smooth surface S to a line integral of the vector field along the boundary curve C of S .
- Stokes' theorem can be written as:

$$\iint_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

- where $\mathbf{F}$ is a vector field that has continuous partial derivatives on an open region that contains S .
- Stokes' theorem is a generalization of the Fundamental Theorem of Calculus, the Fundamental Theorem for Line Integrals, and Green's theorem to higher dimensions.
- Stokes' theorem can be used to evaluate surface integrals without parametrizing the surface S , by finding a related line integral over the curve C .
- Stokes' theorem can also be used to find the flux and divergence of a vector field across a surface S , by choosing $\mathbf{F}$ to be the curl of another vector field.

- Stokes' theorem can also be used to verify the curl identities and the vector identities involving the curl operator.

Applications

- Green's theorem and Stokes' theorem can be applied to various problems in physics, engineering, and geometry, such as calculating work, circulation, flux, curl, divergence, potential, and area.
- Some examples of applications are:
 - Finding the work done by a force field along a closed curve.
 - Finding the circulation of a fluid around a closed curve.
 - Finding the flux of a magnetic field across a surface.
 - Finding the curl of a vector field at a point on a surface.
 - Finding the divergence of a vector field at a point in space.
 - Finding the electric potential of a charge distribution on a surface.
 - Finding the area of a surface or a plane region .
 - Finding the centroid of a plane region.
 - Finding the moment of inertia of a plane region.